

Nutrition in Chronic Kidney Disease

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LIFESTYLE ADVICE: NUTRITION

In the early stages of chronic kidney disease (CKD), the emphasis of disease management is on lifestyle interventions. This includes maintaining or achieving a healthy body weight, ensuring good blood glucose control if diabetes is present, and providing dietary interventions to improve blood pressure (BP) and lipid levels. A healthy diet as set out by the World Health Organization (WHO) encourages:

- A variety of vegetables, fruits, legumes, nuts, and whole grains
- Limiting energy intake from fats (with unsaturated fats being preferable to saturated fats)
- Limiting total energy intake from free sugars
- Reducing salt intake

There has been much discussion about reducing protein in the diet and the extent to which it may help slow progression of CKD. Although most would promote moderating protein intake (0.8–1 g/kg body weight/day), opinions differ regarding the ideal level of protein intake as CKD advances.

Multiple levels of protein restriction have been proposed for patients with CKD stages 3 to 5 not on dialysis, with different recommendations for those with or without diabetes^{1–4} (Table 90.1). However, a Cochrane review outlines the complexity of this evidence and highlights the need for further research before these dietary approaches can be recommended for widespread use.⁵ This therefore remains under discussion.

In people with a solitary kidney (e.g., after nephrectomy) or at high risk for CKD, consider moderating protein to 0.8 to 1.0 g/kg body weight as indicated in the International Society of Nutrition and Metabolism (ISRNM) Commentary.^{3,6} Certain patient populations, such as patients with polycystic kidney disease, appear to benefit less from a low protein diet,¹ but even these patients should avoid high dietary protein intake greater than 1.0 g/kg/day.⁶ A higher protein intake of 1.0 to 1.2 g/kg body weight/day is recommended for dialysis-dependent patients.¹

When advising patients with CKD about their protein requirements, factors to consider include diagnosis, CKD stage, comorbidity, nutritional status, food preferences, motivation, cooking skills, and dietetic resources. Lower levels of protein may be advised for those who are motivated, otherwise well/stable, and able to be closely monitored. Low-protein diets can be difficult to implement, in practice requiring adequate resourcing of specialist renal dieticians to allow clinical judgment to be exercised. Individuals need careful monitoring of nutritional status to help prevent malnutrition while acknowledging patient preferences and adherence.

More recently, a plant-dominant (PLADO) low-protein diet has also been proposed, where more than 50% of protein is sourced from plants with a total daily protein intake in the range

of 0.6 to 0.8 g/kg body weight/day. This diet was designed for people with CKD who do not need dialysis and may be of interest for the future.⁷

The determination of body weight for diet prescription, whether it be ideal body weight, body mass index (BMI), usual or current, or adjusted depends on clinical judgment because adjustment must be considered for people who are severely underweight or obese.¹

Malnutrition: Protein-Energy Wasting

Protein-energy wasting (PEW) is defined as a state of nutritional and metabolic derangements in patients with CKD that may negatively affect nutritional status and lean body mass, leading to frailty.⁸ Inflammation often coexists, and nutrition intervention in isolation may not successfully reverse the loss of skeletal muscle and fat mass.⁹

PEW is common among patients with CKD, especially those undergoing maintenance dialysis,^{10,11} and is associated with increased morbidity and mortality. Several factors contribute to the high incidence of PEW:

- Inadequate dietary protein and energy intake: There is a spontaneous reduction in nutrient intake that parallels the decrease in glomerular filtration rate (GFR) and is largely driven by anorexia.¹² Anorexia in CKD may be caused by impaired taste acuity and diminished olfactory function, medications, autonomic gastroparesis, psychological and socioeconomic factors, and inadequate dialysis. Suboptimal intake can be further compounded by frailty, poverty, advanced age, and multiple acute or chronic comorbidities.
- Inappropriate or overzealous dietary restrictions may have been recommended, which is particularly relevant for patients with end-stage kidney disease (ESKD).^{13,14}
- Protein and amino acid losses occur during dialysis treatment. For those on peritoneal dialysis (PD), these losses may increase during episodes of peritonitis.
- Metabolic acidosis and periods of acute or chronic illnesses may induce protein catabolism. This is mediated in large part through the ubiquitin-proteasome pathway of protein degradation.¹⁵
- Protein catabolic effects that can further compromise patients early after transplantation include large doses of corticosteroids, the stress response to surgery, and delayed graft function.
- Chronic inflammation may contribute to both an increase in nutritional needs and anorexia. Alterations in intestinal microbiota and increased permeability of the intestinal barrier may contribute to the pathogenesis of inflammation (see Chapter 83).¹⁶
- Endocrine disorders, such as insulin resistance (associated with increased muscle breakdown), vitamin D deficiency, and increased parathyroid hormone concentrations have long been considered contributors to protein energy wasting, although the data remain inconclusive regarding the exact mechanisms involved. There is

TABLE 90.1 Guidelines for Levels of Dietary Protein Intake for Chronic Kidney Disease (Stages 3–5) Not on Dialysis

Protein (g/kg BW/day) ^a	Comments
0.8–1.0 g/kg/day	Widely adopted to moderate protein for CKD patients
0.8 g/kg/day	KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease ⁴ (CKD [stages 4–5] patients with or without diabetes)
	KDIGO 2020 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease ² (CKD patients with diabetes)
0.6–0.8 g/kg/day	KDOQI clinical practice guideline for nutrition in CKD:2020 update ¹ (CKD patients with diabetes under close supervision)
	ISRNM Commentary on the KDOQI clinical practice guideline for nutrition in CKD 2020 ³ (suggests a more streamlined target for CKD patients with or without diabetes under close supervision)
0.55–0.6 g/kg/day or 0.28–0.43 g/kg/day (supplemented with essential keto acid/amino acid analogs to meet protein requirements (0.55–0.60 g/kg body weight/day)	KDOQI clinical practice guideline for nutrition in CKD:2020 update ¹ (CKD patients without diabetes who are metabolically stable and under close supervision)

^aDetermination of body weight depends on clinical judgment.¹

BW, Body weight; CKD, chronic kidney disease; KDIGO, Kidney Disease Improving Clinical Outcomes; KDOQI, Kidney Disease Outcomes Quality Initiative; ISRNM, International Society of Nutrition and Metabolism.

some evidence of an association between fibroblast growth factor-23 (a phosphate-regulating hormone) and inflammation in CKD.¹⁷

Obesity Paradox in Chronic Kidney Disease

Although there is a high prevalence of PEW in patients with CKD, which is associated with poorer outcomes, paradoxically, a higher BMI is associated with better survival in patients receiving hemodialysis (HD). This is termed *reverse epidemiology* (see [Chapter 84](#)). This relationship does not hold for morbid obesity in patients receiving HD, which may be used as a predictor of higher mortality.¹

Of note, obesity in some centers can prevent a patient being activated on the transplant list, and support to achieve an acceptable BMI is often needed.

Assessment of Nutritional Status

The measurement of nutritional status does not lend itself to one simple test, and a panel of measures is required. [Table 90.2](#) summarizes some of the methods used for the assessment of nutritional status.

ESTIMATION OF INTAKE

Diet history, recall, and food diaries are the mainstays for estimation of dietary nutrient intake. In dialysis-dependent patients, a gradual decrease in blood urea nitrogen (BUN) and reduced phosphate and potassium levels may indicate a decrease in protein intake, and a low serum cholesterol level may indicate a poor calorie intake.

In HD, the protein catabolic rate (PCR), also known as the protein equivalent of total nitrogen appearance (PNA), can be estimated from interdialytic changes in BUN concentration in serum. The interdialytic appearance of urea is determined by measuring the BUN at the end of one dialysis session and just before the start of the next dialysis session. Among patients who have low BUN, measuring the change in BUN between dialysis sessions distinguishes adequate dialysis as a cause of low BUN from poor intake resulting from inadequate dialysis. The PCR is expressed as g/day. Usually, the PCR is normalized to the patient's body weight (nPCR), which is expressed as g/kg/day.

Assuming that nitrogen excretion equals nitrogen intake in steady state, nPCR can be used to approximate dietary protein intake in the short term but results should be interpreted with caution. The PCR in patients treated with HD is calculated by various urea kinetic modeling

TABLE 90.2 Assessment of Nutritional Status

Area	Assessments
Physical Examination	
Assessment of dietary intake	Diet history, food diaries, appetite assessment questionnaires
Anthropometric measurements	Body weight, height, body mass index Percentage weight change Skinfold thickness Mid arm muscle circumference Waist circumference
Body composition	Bioelectrical impedance DEXA Near-infrared reactance Neutron activation Total body potassium
Biochemical determinations	Serum electrolytes Serum proteins PNA, PCR Serum cholesterol Creatinine index ^a
Nutritional scoring systems	Subjective Global Assessment, Malnutrition Inflammation Score (MIS), Dialysis Malnutrition Score (DMS) Geriatric Nutritional Risk Index (GNRI)
Immunologic assays	Blood lymphocyte count Delayed cutaneous hypersensitivity tests
Functional tests	Grip strength

^aThe creatinine index is measured as the sum of creatinine removed from the body (measured from the creatinine removed in dialysate, ultrafiltrate, and/or urine), any increase in the body creatinine pool, and the creatinine degradation rate. See also www.kidney.org/professionals/kdoqi/pdf/KDOQI2000NutritionGL.pdf.

DEXA, Dual-energy x-ray absorptiometry; PCR, Protein catabolic rate (mathematically identical to PNA); PNA, protein equivalent of total nitrogen appearance.

software programs. (Urea kinetic modeling and adequacy of dialysis are further discussed in [Chapters 99 and 101](#).)

The use of PCR to estimate protein intake has a number of limitations. It may be considered as a complementary tool to assess

nutritional status but should not be interpreted in isolation because it can be influenced by nonnutritional factors.¹

If a computer program is not available, the following simple formula will give a good estimate of the nPCR:

$$\text{nPCR (g/kg/day)} = 0.22 + \frac{(0.036 \times \text{interdialytic rise in BUN [mg/dL]} \times 24)}{\text{interdialytic interval (hours)}}$$

Among patients with residual kidney function, total urea lost in the urine must also be measured from urine collected during the interdialytic period. Using this measurement, the following term is added to the equation for nPCR^{17a}:

$$+ \frac{\text{Urinary urea nitrogen (g)} \times 150}{\text{Interdialytic interval (hours)} \times \text{Weight (kg)}}$$

where the urinary urea nitrogen is all of the urea nitrogen excreted in a urine collection obtained from the end of one dialysis to the beginning of the next (i.e., in the interdialytic interval).

Body Mass Index

BMI (weight [kg]/height² [m²]), also known as the Quetelet index, is the most commonly used parameter for nutritional assessment; however, because weight does not distinguish between muscle and fat mass or acknowledge hydration status, it may not be representative.

Body Composition

A range of techniques can distinguish body compartments based on physical characteristics providing information about nutritional state (body lean tissue and fat content) and hydration. Some are costly, less accessible, and used more for research purposes, and the usefulness of the results depends on the availability of reference data from a population of same age, race, sex, and disease status. A few of the more widely available techniques are mentioned in the following discussion.

Triceps skinfold thickness can be used as an indicator of fat mass, and mid-arm muscle circumference (MAMC) as a surrogate measure of fat-free mass. The midpoint of the upper dominant arm is used because this is the arm less likely to have an arteriovenous fistula (Fig. 90.1).

$$\text{MAMC (cm)} = \text{Mid-arm circumference (cm)} - [3.14 \times \text{Triceps skinfold}] \text{ (cm)}$$

The measurement is taken after dialysis for patients on HD. Although these anthropometric parameters are inexpensive and relatively easy to measure, they are limited by relatively wide ranges of inter- and inpatient variability. An estimation of BMI can be made from the mid-upper arm circumference and there are reference values available, but caution is needed because these are not disease specific. Nevertheless, serial measures over time can be useful to track changes in the same patient when they are used in conjunction with other nutritional indices.

An unintentional weight loss of 5% to 10% would be interpreted as a greater than normal intraindividual variation and an earlier indicator of increased risk of undernutrition, and greater than 10% would be interpreted as clinically significant.¹⁸

Observation of trends using serial measures of bioelectrical impedance are increasingly being used as an adjunct to the day-to-day clinical assessment of hydration status and body composition management of dialysis patients. Assessments are best obtained after dialysis treatment when fluid compartment levels are balanced. Bioelectrical impedance is not routinely available in all facilities and is not without its limitations. Further prospective controlled trials are required to determine the best role for this technology in clinical practice.^{19–21}



Fig. 90.1 Routine Measurement of Skinfold Thickness. The dominant arm, which does not have an arteriovenous fistula or graft, is used in patients with kidney failure.

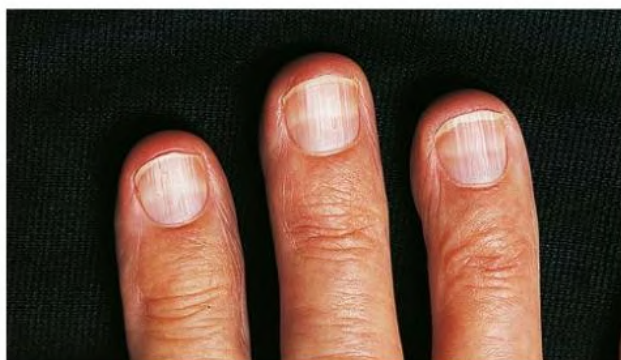


Fig. 90.2 White Nails in Hypoalbuminemia. The white band grew during a transient period of hypoalbuminemia caused by nephrotic syndrome.

Serum Albumin, Prealbumin, and Transferrin

Serum albumin, prealbumin, and transferrin represent the so-called *visceral protein*. Fluid status, impaired liver function, age, and acute inflammatory conditions can affect serum albumin levels; however, despite its relatively long half-life (20 days), albumin remains an important measure of the nutritional status and health of the patient. Clinically, it may be possible to observe the growth of white nails when there has been a transient period of hypoalbuminemia (Fig. 90.2). Other serum protein markers of nutritional status are difficult to interpret because of the influence of factors other than nutrition. Serum transferrin is linked to body iron stores and may be altered with changes in iron status. Serum prealbumin (transthyretin) levels can be increased by CKD because of impaired metabolism in the kidney but also decline rapidly during episodes of acute inflammation.

Tools to Diagnose Protein-Energy Wasting and Assess Nutritional Status

Given the low specificity and sensitivity of many of the anthropometric and biochemical markers, a range of measurements is needed to assess nutritional status in addition to the evaluation of the subjective well-being of the patient (Table 90.3).

Four main established categories are recommended for the screening for PEW: serum biochemistry, body mass, muscle mass, and dietary intake.⁸ Subjective global assessment (SGA) is a reliable nutritional

TABLE 90.3 Some Indices of Malnutrition

Assessment	Indices
Biochemical parameters	Serum albumin below the normal range Serum prealbumin < 300 mg/L (30 mg/dL) (for maintenance dialysis patients only because levels may vary according to GFR level for CKD stages 2–5) Low predialysis serum creatinine, phosphate, potassium, urea levels Serum total cholesterol < 150 mg/dL (3.8 mmol/L) Low creatinine index Low PNA, PCR
Anthropometric parameters	Continuous decline in weight, skinfold thickness, mid-arm muscle circumference <i>WHO classification for BMI:</i> <18.5 kg/m ² underweight (<20 kg/m ² for some classification systems; note that ISRNM suggests a lower threshold of 23 kg/m ²) ⁷ 18.5–24.9 kg/m ² normal weight 25–29.9 kg/m ² overweight >25 kg/m ² obese Body weight < 90% of ideal Abnormal muscle strength

GFR, Glomerular filtration rate; ISRNM, International Society of Renal Nutrition and Metabolism; PCR, protein catabolic rate; PNA, protein equivalent of total nitrogen appearance; WHO, World Health Organization.

assessment tool for patients on dialysis.²² Questions about recent changes in nutrient intake are used alongside simple observations of the patient's body weight and muscle mass to assess the nutritional status of the individual; patients are classified as well nourished, mild to moderately malnourished, or severely malnourished. Figs. 90.3 and 90.4 show muscle wasting in a patient on HD who was classified by SGA as severely malnourished. The Kidney Disease Outcomes Quality Initiative (KDOQI) recommends the use of the 7-point SGA for assessing nutritional status in adults on dialysis.¹ The dialysis malnutrition score (DMS) and the malnutrition inflammation score (MIS), which are fully quantitative adaptations of the SGA, as well as the International Society of Renal Nutrition and Metabolism (ISRNM) PEW definition criteria and the geriatric nutritional risk index, are alternative tools used by some centers and continue to be evaluated and refined.

Nutritional Guidelines, Monitoring, and Treatment

Renal dietitians, or their equivalent in different regions of the world, have the unique knowledge and skills to nutritionally assess, tailor advice, and prioritize goals according to the patient's clinical condition. The renal dietitian is trained in behavioral methods to help reduce barriers, can impart practical information, and may have an extended role, such as with the management of CKD-related bone disease. Latest guidance from KDOQI recommends that a registered dietitian, nutritionist, or regional equivalent collaborate closely with the multiprofessional team to help tailor nutrition advice to patients with CKD.¹

Guidelines are useful, but it is important that dietary restrictions are not unnecessarily imposed and that advice is tailored to the individual and altered as circumstances dictate. Table 90.4 summarizes the nutritional recommendations for CKD.

Historically, dietary advice for patients with CKD has focused on individual nutrients, such as phosphorus, potassium, salt, and protein, which can be complex, challenging to adhere to, and may lead to unbalanced diets. Consequently, particular dietary patterns, including the Mediterranean diet, the Dietary Approach to Stop Hypertension (DASH), and plant-based diets that are high in fruit and vegetables, are increasingly of interest because they are linked with reduced chronic cardiovascular (CV) disease (CVD) and mortality risk in the healthy



Fig. 90.3 Severely malnourished hemodialysis patient.

population.²³ A recent meta-analysis looked at seven studies where healthy dietary patterns (generally higher in fruit, vegetables, fish, legumes, whole grain, and fiber and lower in red meat, salt, and refined sugars) were consumed by adults with CKD and demonstrated that these eating patterns were consistently associated with lower mortality.²⁴ Although traditionally there has been fear of hyperkalemia from plant-based foods, recent studies suggest that a higher intake of dietary fibers from more plant-based sources can enhance bowel movement and gastrointestinal (GI) potassium loss and have an alkalizing affect, which may reduce hyperkalemia.⁶

The safety and acceptability of various diet patterns must be determined on an individual basis in the advanced stages of CKD, especially with regard to serum potassium control and adequacy of protein and energy. More large-scale clinical trials implementing DASH/Mediterranean style and plant-based diets, including PLADO diet, in people with CKD are needed.

Monitoring of patients with CKD involves a combination of nutritional assessment, noting relevant biochemistry (potassium, phosphate, and lipids), checking dialysis adequacy, and observing fluid status. The challenge comes in balancing the potential dietary restrictions of CKD



Fig. 90.4 Severely Malnourished Hemodialysis Patient. There is marked wasting of the quadriceps and calf muscles. In addition, note skin lesions from scratching because of uremic pruritus.

against the risk for compromising nutritional status. When anorexia is present, compromising food intake, nutrient intake may be maximized by one or more of the methods discussed in the following sections.

Enteral Supplementation

If food fortification advice is insufficient, supplements, in the form of high-protein, high-calorie drinks, powders, and puddings, should be considered when there are signs of PEW or its high-risk factors, especially in dialysis patients.¹ Enteral tube feeding is an option if nutrient intake cannot be increased sufficiently using oral supplements.¹ Kidney-specific tube feeds and supplements are available that are lower in fluid and electrolytes. A systematic review suggested that enteral multnutrient support increases serum albumin concentration and improves total dietary intake in patients receiving maintenance dialysis.²⁵ In addition, where maintenance HD patients with albumin levels of up to 3.5 g/dL received intradialytic oral nutritional supplements in a large retrospective matched-cohort study, despite limitations, significantly better survival was seen in the supplemented group versus similarly matched patient controls.^{26,27}

Supplementation of Intradialytic Parenteral Nutrition and Dialysate Fluids

The GI route is the preferred choice for nutritional supplementation; however, intradialytic parenteral nutrition (IDPN), where a concentrated hypertonic solution is infused into the venous blood line three times weekly over approximately 4 hours during HD treatments, has been used to provide intensive parenteral nutrient therapy for patients who cannot tolerate oral or enteral administration of nutrients. IDPN typically provides 800 to 1200 kcal three times weekly, in the form of glucose and fat emulsion and 30 to 60 g of protein, and so will be only supplementary rather than providing full nutritional needs.

A systematic review concluded that there was insufficient evidence to demonstrate either a net benefit or a net harm associated with

providing IDPN to malnourished HD patients,²⁸ but a more recent RCT demonstrated IDPN therapy increased prealbumin levels and was superior to nutrition counseling after 16 weeks.²⁹ Further clinical research is needed in this area, although the high cost of the therapy is a barrier to performing adequately powered clinical trials.

Intraperitoneal amino acids (IPAAAs) can be used in PD. A 1.1% amino acid solution is substituted for glucose in PD fluid, and about 80% of the amino acids are absorbed in a 4-hour period.³⁰ Because the long-term effects of IPAAAs on nutritional status and clinical outcomes are not known, the solution is often used to reduce glucose exposure.

Expert opinion on the use of these approaches is inconsistent. KDOQI has suggested that IPAAAs (for PD) or IDPN (for HD) should be considered for patients who are unable to tolerate adequate oral supplements or tube feeding and have evidence of protein or energy malnutrition.¹ The European Society of Enteral and Parenteral Nutrition (ESPEN) has provided criteria for use of IDPN.³¹ Conversely, the American Society for Parenteral and Enteral Nutrition (ASPEN) recommends that IDPN not be used as a nutritional supplement in malnourished HD patients because studies so far are not convincing.³²

Appetite Stimulants

Despite megesterol acetate, a progesterone derivative, improving appetite moderately in HD patients in small studies, it can have adverse effects, and larger trials are required before recommendations can be made for CKD patients.³³

In some countries, mirtazapine, an antidepressant with the side effect of increasing appetite, and cannabis-based agents can be considered, but research data in their support are insufficient. It is important to note that the licensing and restrictions of these drugs/agents plus ethical considerations are likely to vary between countries, which will affect their usage and suitability for recommendation.

More studies are also required for ghrelin, an orexigenic hormone, and melanocortin-receptor antagonists.

Vitamins, Minerals, and Trace Elements

Vitamin, mineral, and trace element abnormalities in CKD are related to dietary restriction, dialysate losses, and the necessity of intact kidney function for normal metabolism; however, the dietary requirements for patients with CKD are not clear-cut.

Protein and potassium restrictions can lead to inadequate intakes of pyridoxine, vitamin B₁₂, folic acid, vitamin C, iron, and zinc. The use of recombinant human erythropoietin may increase the requirement for iron and folic acid (see [Chapter 86](#)).

Elevated total homocysteine levels have been identified as a risk factor for CVD in the general population. Homocysteine levels in the blood are influenced by blood levels of folic acid; however, evidence from randomized trials does not support supplementation in secondary prevention. Hyperhomocysteinemia is common in patients with ESKD. As observed in those with normal kidney function, lowering homocysteine levels with folic acid, vitamin B₁₂, and pyridoxine supplements had no effect on mortality or risk of CV events in dialysis patients.³⁴

A variety of available vitamin preparations contain a recommended water-soluble vitamin profile for dialysis patients. In the absence of firm guidance, it is prudent to have a low threshold for commencing such a water-soluble vitamin preparation. The European Best Practice (www.european-renal-best-practice.org) on nutrition gives opinion-based recommendations for patients on HD³⁵ and KDOQI give a statement of guidance:

In adults with CKD 3 to 5D or post transplantation, it is reasonable for the registered dietician, nutritionist, or interventional equivalent, in close collaboration with a physician

TABLE 90.4 Nutritional Recommendations in Chronic Kidney Disease^a

Daily Intake	CKD Not on Dialysis	Hemodialysis	Peritoneal Dialysis
Protein (g/kg BW/day)	0.6–1.0 Level depends on the view of the nephrologist. 1.0 for nephrotic syndrome (see Table 90.1)	1.0–1.2 ¹ Recommendations are in conjunction with an adequate energy intake. Requirements may be higher during illness because of multiple comorbidities or during acute periods of infection, including peritonitis.	1.0–1.2 ¹
Energy (kcal/kg BW)	25–35 ¹	25–35 ¹	25–35 ¹
Sodium (mmol)	<100 ¹ (more if salt wasting)	<100 ¹	<100 ¹
Potassium	Reduce if hyperkalemic If hyperkalemic, patients will be advised to moderate intake of certain foods (e.g., some fruits and vegetables) and given information about cooking methods.	Reduce if hyperkalemic	Reduce if hyperkalemic; potassium restriction is generally not required. May need to increase potassium intake if hypokalemic.
Phosphorous	Adjust dietary phosphate intake to maintain serum phosphate levels in the normal range. Patients will be advised to moderate intake of certain foods (e.g., dairy, offal, some shellfish) as well as processed foods with high content of added phosphates, and given information about the timing of binders with high-phosphorus meals and snacks.		

^aRecommendations are for typical patients but always should be individualized on the basis of clinical, biochemical, and anthropometric indices. BW, Body weight; CKD, chronic kidney disease.

Determination of body weight depends on clinical judgment.¹

or physician assistant, to assess dietary vitamin intake periodically and to consider multivitamin supplementation for individuals with inadequate vitamin intake. In adults with CKD 5D who exhibit inadequate dietary intake for sustained periods of time, it is reasonable to consider supplementation with multivitamins, including all the water-soluble vitamins, and essential trace elements to prevent or treat micronutrient deficiencies.¹

High-dose vitamin C supplements should not be taken by people with CKD because of the increased risk for secondary tissue oxalate deposition.

A review on fat-soluble vitamins in advanced CKD concluded that supplementation with vitamin A is generally not recommended (unless a patient is receiving total parenteral nutrition) because deficiencies are rare, dialysis losses are minimal, and accumulation leading to toxicity can occur.³⁶ Vitamin E may have antioxidant properties and beneficial effects for patients with CKD,³⁷ but appropriately powered studies with longer follow-up are needed for confirmation.

Vitamin K is a cofactor for the enzyme γ -glutamyl carboxylase and may inhibit vascular calcification (see [Chapters 84 and 88](#)). Evidence suggests most dialysis-dependent patients have subclinical vitamin K deficiency, and there is no known toxicity. Research is ongoing to determine whether vitamin K supplementation may prevent vascular calcification and reduce mortality in patients with advanced CKD.³⁸ KDOQI suggests that patients receiving anticoagulant medicines known to inhibit vitamin K activity (e.g., warfarin compounds) should not receive vitamin K supplements.¹

Guidelines for vitamin D, phosphorus, and calcium are summarized in the Kidney Disease: Improving Global Outcomes (KDIGO) guidelines (see [Chapter 88](#)).

Recommendations for sodium, potassium, and phosphorus are shown in [Table 90.4](#).

Gut-Targeted Therapeutics

There is accumulating evidence that the GI tract may be a major source of chronic inflammation in CKD. It is hypothesized that altered diets (low potassium, phosphorus, and fiber) may affect the gut microbiome, resulting in overgrowth of bacteria that produce uremic toxins and a leaky epithelial barrier, allowing toxins into the circulation, whereas plant-based diets may have a salutary effect.

Prebiotic and probiotic formulations may lower serum levels of uremic toxins, but more trials investigating gut-targeted therapeutics are needed before they can be recommended for use in clinical practice.

Metabolic Acidosis

Although some trials have shown no detrimental effect of mild metabolic acidosis, many others have reported that normalization of serum bicarbonate concentration (see [Chapter 13](#)) is beneficial for nutritional status and bone metabolism. Current guidelines recommend the correction of acidosis in dialysis-dependent patients^{9,39,40} and KDOQI states that in adults with CKD 3 to 5D, it is reasonable to maintain serum bicarbonate levels at 24 to 26 mmol/L.¹

Exercise

There is a large body of research on exercise training for adults with CKD that shows significant benefits on physical fitness, CV parameters, well-being, and nutritional markers.⁴¹ Trials are ongoing, and the future is likely to hold more guidance in this area for clinical practice.

SELF-ASSESSMENT QUESTIONS

1. A 60-year-old woman began hemodialysis 6 years ago. At a recent outpatient appointment, she reported breathlessness. Predialysis BP was 150/80 mm Hg.
 - Dialysis: 4 hours three times per week
 - Target weight: Stable for last 2 years at 55 kg

- Height: 1.47 m
- Body mass index (BMI): 25 kg/m²
- Intradialytic weight gain average: 0.9 kg
- Passes small amounts of urine

Monthly laboratory testing reveals blood results as follows:

	Creatinine (mg/dL)	Urea (mg/dL)	Phosphate (mg/dL)	Potassium (mmol/L)	Albumin (g/dL)	Cholesterol (mg/dL)	Hemoglobin (g/dL)
Predialysis	5.3	27	3.16	4.5	3.5	100	13.85
Postdialysis	1.54	7.86	1.30	2.9			

What would your next course of action be?

- A. Reduce the dialysis time because she is receiving more dialysis than required.
 - B. Investigate dietary intake, in particular protein and calorie intake, and assess target weight.
 - C. Increase target weight.
 - D. Start diuretics and fluid restriction.
2. A 38-year-old White man with progressive CKD is feeling tired and reporting weight loss, nausea, and a metallic taste in the mouth. His BP is 156/90 mm Hg, and urinalysis reveals +++ protein and traces of blood. There are no signs of peripheral edema. He trains at the gym three times per week.
- Weight: 78 kg
 - Height: 1.7 m
 - BMI: 27 kg/m²
 - Amlodipine 5 mg/day begun
 - Sodium bicarbonate begun
 - Diet history reveals an average daily protein intake of 120 g, an approximate calorie intake of 2600 kcal, and sodium intake of 190 mmol/day. Laboratory findings are as follows:

What would your next course of action be?

- A. Put the patient on a reduced-calorie diet.
 - B. Start dialysis.
 - C. Moderate protein and salt intake.
 - D. Start a low-potassium diet.
3. A 59-year-old man with ESKD resulting from type 1 diabetes has been treated with continuous ambulatory peritoneal dialysis for 12 months. He reports reduced appetite, occasional nausea, and feeling full only midway through modest-sized meals; he has lost 3.3 kg over the previous 2 months. He weighs 79.2 kg and has a BMI of 34 kg/m². He dialyzes with four exchanges of volume, 2 L each. Measurement of dialysis clearances shows a weekly Kt/V for urea of 1.8 (target minimum 1.7). What would your next course of action be?
- A. Reduce dialysis fluid volumes in case they are contributing to his gastrointestinal symptoms.
 - B. Advise weight reduction because BMI may be contributing to feelings of fullness.
 - C. Start amino acid-containing peritoneal dialysate.
 - D. Start a prokinetic agent.

Glomerular filtration rate (mL/min)	Creatinine (mg/dL)	Urea (mg/dL)	Phosphate (mg/dL)	Calcium (mg/dL)	Potassium (mmol/L)	Bicarbonate (mEq/L)
13	5.1	74	4.96	10.02	5.1	16

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